BIO 005 Human Physiology · Week 8 notes · Dr. Sharilyn Rennie

The autonomic nervous system

The written notes for the autonomic nervous system walkthrough. Week 8 competencies 1 to 5.

To print: these notes as a two-column PDF. Already printed the Week 8 notes? You do not need to print these again.

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Step by step: where the signal travels

In your textbook: Silverthorn Chapter 12, the parts on the autonomic division and the somatic motor division.

A somatic motor command, step by step

1. The cell body of the somatic motor neuron sits in the ventral horn of the spinal cord (or in the brainstem).
2. Its single, myelinated axon leaves through the ventral root and runs all the way to a skeletal muscle. There is no synapse between the cord and the muscle.
3. At the neuromuscular junction it releases acetylcholine (ACh), which binds nicotinic receptors on the muscle fiber.
4. The muscle fiber depolarizes and contracts. A somatic motor neuron can only excite its muscle.

A parasympathetic signal to the heart, step by step

1. The preganglionic neuron's cell body sits in the brainstem.
2. Its long axon travels in the vagus nerve (cranial nerve X) to a ganglion on the heart.
3. In the ganglion it releases ACh onto nicotinic receptors on the postganglionic neuron, which depolarizes and fires.
4. The short postganglionic axon reaches the SA node and releases ACh from its varicosities.
5. ACh binds M2 muscarinic receptors on the pacemaker cells. K+ channels open and less Ca2+ enters, so the cells take longer to reach threshold.
6. The heart rate slows.
7. Acetylcholinesterase breaks down the ACh, ending the signal.

A sympathetic signal to an arteriole, step by step

1. The preganglionic neuron's cell body sits in the lateral horn of the thoracic or lumbar spinal cord.
2. Its short axon leaves through the ventral root and enters a ganglion of the sympathetic chain beside the cord.
3. It releases ACh onto nicotinic receptors on postganglionic neurons. It can branch up and down the chain and reach many of them.
4. The long postganglionic axon runs to the arteriole and swells into varicosities along the vessel wall.
5. The varicosities release norepinephrine, which binds alpha-1 receptors on the smooth muscle.
6. IP3 and Ca2+ rise inside the smooth muscle cells, the muscle contracts, and the arteriole narrows.
7. Norepinephrine diffuses away or is taken back into the varicosity and broken down by monoamine oxidase (MAO).

A sympathetic signal to a sweat gland, step by step

1. The preganglionic neuron leaves the thoracic or lumbar cord and synapses in a chain ganglion, releasing ACh onto nicotinic receptors.
2. The postganglionic neuron, a sympathetic cholinergic neuron, runs to the sweat gland.
3. It releases ACh, not norepinephrine, onto muscarinic receptors on the gland, and the gland secretes sweat.

A sympathetic signal through the adrenal medulla, step by step

1. The preganglionic neuron leaves the thoracic cord and runs straight to the adrenal medulla, with no ganglion on the way.
2. It releases ACh onto nicotinic receptors on chromaffin cells, which are modified postganglionic neurons without axons.
3. The chromaffin cells secrete epinephrine, with some norepinephrine, into the blood.
4. Epinephrine travels as a neurohormone to adrenergic receptors throughout the body, for example beta-2 receptors on bronchiole smooth muscle, which relaxes and widens the airway.

Heart rate with each breath, step by step

1. You breathe in.
2. The brainstem briefly turns down vagal (parasympathetic) firing to the SA node.
3. Less ACh reaches the M2 receptors, and the heart speeds up within about a beat.
4. You breathe out, vagal firing returns, and the heart slows.
5. This rise and fall is respiratory sinus arrhythmia, and its size reflects vagal tone.

Two efferent pathways

In your textbook: Silverthorn Chapter 12, the parts on autonomic pathways with two efferent neurons and on the somatic motor division.

The efferent (output) side of the peripheral nervous system has two divisions. The somatic motor division controls skeletal muscle. The autonomic division, also called the visceral nervous system, controls cardiac muscle, smooth muscle, many glands, and some adipose (fat) tissue.

	Somatic motor	Autonomic
Neurons from the CNS to the target	One	Two in series: preganglionic and postganglionic
Synapse outside the CNS	None before the target	In an autonomic ganglion
Effector	Skeletal muscle	Cardiac muscle, smooth muscle, glands, some adipose tissue
Effect on the target	Excitatory only	Excitatory or inhibitory
Transmitter at the target	ACh on nicotinic receptors	ACh on muscarinic receptors, or norepinephrine on adrenergic receptors
If the nerve is cut	The muscle is paralyzed and wastes away	The heart and many smooth muscles keep working on their own

Preganglionic neuron

The first autonomic neuron. Its cell body is in the brain or spinal cord, and its axon ends in an autonomic ganglion.

Autonomic ganglion

A cluster of nerve cell bodies outside the CNS, where the preganglionic neuron synapses on postganglionic neurons. One preganglionic neuron can reach many postganglionic neurons there (divergence), and some ganglia work as small integrating centers.

Postganglionic neuron

The second autonomic neuron. Its cell body is in the ganglion, and its axon runs to the target tissue.

- The heart's pacemaker cells start each beat on their own, so autonomic nerves speed up or slow down a heart that is already beating. A transplanted heart, with its nerves cut, still beats.
- Skeletal muscle has no rhythm of its own, so every contraction starts with a somatic motor neuron.

Sympathetic and parasympathetic

In your textbook: Silverthorn Chapter 12, the part on where the sympathetic and parasympathetic branches originate.

	Sympathetic	Parasympathetic
Where preganglionic neurons leave the CNS	Thoracic and lumbar spinal cord (thoracolumbar)	Brainstem, in cranial nerves, and sacral spinal cord (craniosacral)
Where the ganglia are	Mostly in the sympathetic chain ganglia, two chains beside the spinal cord; a few collateral ganglia along the aorta	On or in the walls of the target organs
Preganglionic axon	Short	Long
Postganglionic axon	Long	Short
Spread of one signal	Wide: one preganglionic neuron branches through the chain to many postganglionic neurons	Local: one organ at a time
Heart rate	Increases, and beats are stronger	Decreases
Gut movement (motility) and secretion	Decrease	Increase
Situation it suits	Fight-or-flight	Rest-and-digest

- The vagus nerve, cranial nerve X, carries parasympathetic fibers to the organs of the chest and most of the abdomen.
- Fight-or-flight and rest-and-digest are simplified labels. Both divisions are active most of the time, and the body usually shifts the balance between them rather than switching one off.
- In a crisis the sympathetic division gives priority to the heart and skeletal muscle, and digestion waits.

Transmitters and receptors

In your textbook: Silverthorn Chapter 12, the parts on autonomic neurotransmitters, autonomic receptor subtypes and the adrenal medulla.

Where	Transmitter	Receptor
Every autonomic ganglion, both divisions	ACh	Nicotinic (a ligand-gated channel)
Parasympathetic postganglionic ending	ACh	Muscarinic (G protein-coupled)
Most sympathetic postganglionic endings	Norepinephrine	Alpha or beta adrenergic (G protein-coupled)
Sympathetic ending on most sweat glands	ACh	Muscarinic
Adrenal medulla (chromaffin cells)	ACh from the preganglionic neuron	Nicotinic; the cells then secrete epinephrine into the blood

Receptor	Responds best to	Example location	Response
Nicotinic (ganglionic)	ACh	Postganglionic neurons in all autonomic ganglia; adrenal medulla	Na+ in, more than K+ out: the neuron depolarizes and fires
Muscarinic M2	ACh	SA node of the heart	K+ channels open and Ca2+ entry falls: heart rate slows
Muscarinic	ACh	Most parasympathetic targets; most sweat glands	Depends on the tissue, through second messengers
Alpha-1	Norepinephrine more than epinephrine	Smooth muscle of most blood vessels	IP3 and Ca2+ rise: the muscle contracts and the vessel narrows
Beta-1	Norepinephrine and epinephrine about equally	Heart	cAMP rises: faster and stronger beats
Beta-2	Epinephrine more than norepinephrine	Bronchiole smooth muscle; some blood vessels, such as those in skeletal muscle	cAMP rises: the muscle relaxes, so the airway or vessel widens

Varicosities

Swellings along the end of a postganglionic axon, filled with vesicles. Transmitter diffuses from them to any nearby cells with receptors, so one axon affects many cells.

Cholinergic

Releasing or responding to ACh.

Adrenergic

Releasing or responding to norepinephrine or epinephrine.

Chromaffin cells

The cells of the adrenal medulla: modified sympathetic postganglionic neurons with no axons, which secrete epinephrine (and some norepinephrine) into the blood as a neurohormone.

- The nicotinic receptors in autonomic ganglia are a different subtype from those at the neuromuscular junction, so a drug can block one without the other.
- ACh is broken down by acetylcholinesterase. Norepinephrine diffuses away or is taken back into the varicosity and broken down by monoamine oxidase (MAO).
- Bronchiole smooth muscle has few sympathetic endings, so its beta-2 receptors respond mainly to epinephrine from the adrenal medulla.
- The response is set by the receptor, not by the molecule: the same adrenergic signal narrows an arteriole (alpha-1) and widens a bronchiole (beta-2).
- The adrenal medulla is described as a modified sympathetic ganglion.

Dual innervation and tone

In your textbook: Silverthorn Chapter 12, the parts on antagonistic control and tonic control by the autonomic division.

Dual innervation

An organ receives fibers from both divisions.

Antagonistic control

The two divisions have opposite effects on the same organ, as on the heart rate.

Autonomic tone

A steady, low rate of firing at rest. Turning it up or down moves the organ in either direction.

- With no nerves acting on it, the SA node fires at about 100 beats per minute (roughly 90 to 110). The resting heart beats at about 70.
- Cutting or blocking the vagus nerve raises the heart rate to the SA node's own rate, about 100. Cutting the sympathetic supply at rest slows the heart only a little. So at rest the parasympathetic division dominates the heart.
- Removing one division from a dually innervated organ leaves the other unopposed, so the organ moves the way the remaining division pushes it.
- During exercise, vagal tone falls and sympathetic firing rises, and the heart rate climbs above 100.
- Most blood vessels and the sweat glands receive only sympathetic neurons. Steady sympathetic tone keeps arteriole smooth muscle partly contracted. More firing narrows the vessel (vasoconstriction); less firing than at rest widens it (vasodilation).
- Losing sympathetic tone to the vessels, as after some injuries high in the spinal cord, widens them and can drop blood pressure.

Heart rate with each breath (lab: autonomic testing)

- Heart rate rises during inspiration and falls during expiration. This is respiratory sinus arrhythmia, and it is normal, usually largest in young, healthy people.
- It comes from the vagus nerve: vagal activity is turned down with each breath in and returns with each breath out.
- Only the vagus acts fast enough to change the rate within one breath. ACh at the SA node takes effect within about a second and wears off as quickly, while sympathetic effects through cAMP build and fade over several seconds.
- A large beat-to-beat variation with breathing means strong vagal tone. You measure it in lab as heart rate variability.

Autonomic drugs

In your textbook: Silverthorn Chapter 12, the part on autonomic agonists and antagonists.

A drug that blocks a receptor acts on every organ that uses that receptor. Knowing which receptor dominates in each organ lets you predict both the effect a drug is given for and its side effects.

Drug	Blocks	Wanted effect	Side effects to predict
Propranolol (nonselective beta blocker)	Beta-1 and beta-2	Slower, weaker heartbeat; lower blood pressure	Airway narrowing and wheezing, dangerous in asthma (beta-2 on bronchiole smooth muscle)
Metoprolol (beta-1 selective, cardioselective)	Mainly beta-1	Slower, weaker heartbeat; lower blood pressure	Airways mostly spared at usual doses; selectivity is not complete at high doses
Atropine (muscarinic antagonist)	Muscarinic	Faster heart rate in a dangerously slow heart	Dilated pupils and blurred near vision, dry mouth, slower gut and constipation, less sweating with hot, dry skin

- Beta blockers treat high blood pressure and some heart conditions by blocking beta-1 receptors on the heart, so it pumps less blood each minute.
- Atropine blocks the parasympathetic division at every target, and the sympathetic cholinergic neurons to the sweat glands too. It speeds the heart by blocking the vagal tone at the SA node.
- Near vision blurs with atropine because the muscle that focuses the lens for close work is under parasympathetic control.
- Drugs are often designed to reach one organ. An inhaled muscarinic blocker widens the airways in some lung diseases, because breathing it in keeps most of the drug in the lungs.

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How hormones work

The written notes for the how hormones work walkthrough. Week 8 competencies 7 to 11.

To print: these notes as a two-column PDF. Already printed the Week 8 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: from gland to target cell

In your textbook: Silverthorn Chapter 8, the parts on the classification of hormones and on control of hormone release.

A peptide hormone, step by step (insulin as the example)

1. Ribosomes on the rough ER build a long, inactive chain, the preprohormone.
2. Inside the ER it is trimmed to a prohormone.
3. The Golgi apparatus packages the prohormone into secretory vesicles, where enzymes cut it into the active hormone. The vesicles are stored in the cell.
4. A trigger arrives. For insulin it is a rise in blood glucose, which the beta cells of the pancreas sense directly.
5. The vesicles fuse with the cell membrane and release the hormone by exocytosis.
6. The hormone dissolves in plasma and travels free. Insulin's half life is about 5 minutes.
7. Free hormone leaves the capillary and binds a receptor on the surface of the target cell.
8. The receptor acts through a second messenger or an enzyme, changing proteins the cell already has, within seconds to minutes.

A steroid hormone, step by step (cortisol as the example)

1. A trigger arrives at the cell.
2. Enzymes in the smooth ER and mitochondria convert cholesterol into the hormone, one step at a time. Nothing is stored.
3. The hormone diffuses out through the cell membrane as soon as it is made.
4. In the blood, most molecules bind carrier proteins. For cortisol, only about 5 to 10 percent is free.
5. Only the free hormone leaves the capillary. As it does, bound molecules let go of their carriers to replace it.
6. Bound hormone is protected from breakdown and filtering, so the half life is longer: about an hour for cortisol.
7. The free hormone diffuses into the target cell and binds a receptor in the cytoplasm or nucleus.
8. The hormone and receptor turn a gene on or off. The cell makes messenger RNA and a new protein, so the response takes 90 minutes or more.

Three triggers for release, step by step

1. Humoral: blood Ca2+ falls, parathyroid cells sense it and release PTH, PTH acts on bone and kidney, blood Ca2+ rises, and PTH release slows (negative feedback).
2. Neural: a sympathetic neuron from the spinal cord releases acetylcholine onto chromaffin cells in the adrenal medulla, which release epinephrine (and some norepinephrine) into the blood within seconds.
3. Hormonal: TSH from the anterior pituitary travels in the blood to the thyroid, which releases thyroid hormone (T3 and T4). The next walkthrough covers where TSH comes from.

Three kinds of hormone

In your textbook: Silverthorn Chapter 8, the parts on hormones and on the classification of hormones.

Dr. Rennie's videos for this part: Why the body uses endocrine signals, The endocrine signaling pathway, Primary and secondary endocrine glands, The classes of hormones, Surface and intracellular receptors, Intracellular receptors, Hormone signal transduction.

Hormone

A chemical made by specialized cells, released into the blood, and carried to distant target cells, where it acts at very low concentrations.

Endocrine cell

A cell that makes and releases a hormone. Many sit together in endocrine glands, such as the thyroid, adrenal glands and pancreas.

Target cell

A cell with a receptor for the hormone. The blood carries a hormone past almost every cell, but only target cells respond.

	Peptide	Steroid	Amine
Made from	Amino acids, as a preprohormone, then a prohormone	Cholesterol	One amino acid: tyrosine (catecholamines, thyroid hormones) or tryptophan (melatonin)
Where it is made	Rough ER, packaged by the Golgi	Smooth ER and mitochondria	Cytoplasm (catecholamines); thyroid follicle (thyroid hormones)
Stored?	Yes, in secretory vesicles, made ahead	No, made when needed	Catecholamines: yes, in vesicles. Thyroid hormones: stored in the thyroid gland, not in vesicles
How it leaves the cell	Exocytosis	Diffuses through the membrane	Catecholamines: exocytosis. Thyroid hormones: diffuse out
Dissolves in	Water (lipophobic)	Lipid (lipophilic)	Catecholamines: water. Thyroid hormones: lipid
Receptor	On the cell surface	In the cytoplasm or nucleus	Catecholamines: surface. Thyroid hormones: nucleus
Response	Seconds to minutes, by changing existing proteins	90 minutes or more, by changing genes and making new protein	Catecholamines like peptides; thyroid hormones like steroids
Examples	Insulin, glucagon, parathyroid hormone	Cortisol, aldosterone, estrogen, progesterone, testosterone	Epinephrine, norepinephrine, dopamine, thyroid hormones, melatonin

- Most hormones are peptides.
- Surface receptors for peptides mostly pass the message inside through second messengers such as cAMP, or act as enzymes, as you saw in Week 2.
- Some steroids also bind membrane receptors and cause fast effects that do not involve genes, but the classic steroid action is the slow one through genes.
- Amines do not behave as one class. What decides how each one behaves is the molecule itself: catecholamines dissolve in water, and thyroid hormones (two tyrosines with iodine attached) dissolve in lipid.

Traveling in the blood

In your textbook: Silverthorn Chapter 8, the parts on the classification of hormones and on hormone transport and half life.

Dr. Rennie's videos for this part: Hormone transport: free and bound hormone, How long hormone effects last and how far they reach.

- Peptides and catecholamines dissolve in plasma and travel free.
- Steroids and thyroid hormones dissolve poorly in water, so most of their molecules travel bound to carrier proteins. Some carriers are general, such as albumin; some carry one hormone, such as thyroxine-binding globulin.
- Only free hormone can leave the capillary and bind a receptor. The carrier is too large to leave the blood.
- Bound hormone is a reserve. As free hormone leaves the blood, bound molecules let go of their carriers to replace it.
- Free fraction: about 5 to 10 percent of plasma cortisol; well under 1 percent of thyroxine.

Half life

The time it takes for half of a hormone to disappear from the blood.

How hormones are cleared

Enzymes in the blood and tissues break them down, the liver changes them into forms the body can get rid of, and the kidneys filter free hormone into the urine.

Hormone	Carried	Approximate half life
Insulin (peptide)	Free	About 5 minutes
Cortisol (steroid)	Mostly bound	About an hour
Thyroxine (thyroid hormone)	Almost all bound	About a week

- Binding protects a hormone from the enzymes that break it down, and makes it too large for the kidneys to filter, so bound hormones last longer.
- A short half life lets the body change a hormone's level quickly; a long half life gives a steady level that changes slowly.

More carrier protein, step by step

1. The carrier protein level rises, as thyroxine-binding globulin does in pregnancy or with estrogen.
2. The new carriers bind free hormone, so free hormone falls.
3. Feedback control responds to free hormone, so the gland releases more.
4. Free hormone returns to normal; total hormone is now higher.
5. The person feels well. When carrier levels may have changed, the useful lab number is the free hormone level, not the total.

Receptors decide the response

In your textbook: Silverthorn Chapter 8, the part on hormones and their receptors, with the review of receptors in Chapter 6.

Dr. Rennie's videos for this part: Receptor specificity and a hormone's effects, Receptor regulation and sensitivity to hormones, Receptor regulation and adaptation.

The response to a hormone depends on which receptor the target cell has and which pathway that receptor turns on. The same hormone can cause opposite responses in different tissues.

Tissue	Receptor	What epinephrine does
Smooth muscle of the airways	Beta-2	Relaxes, so the airway widens
Smooth muscle of small arteries in the skin and gut	Alpha-1	Contracts, so the artery narrows
Heart muscle	Beta-1	Beats faster and more forcefully
Small arteries in skeletal muscle	Beta-2	Relaxes, so the artery widens

- Beta receptors raise cAMP. Alpha-1 receptors raise Ca2+ inside smooth muscle cells.
- Together these shift blood and air toward working muscles during exercise or danger.

Down-regulation

Long-lasting high hormone, over hours to days, makes the target cell remove receptors. The dose-response curve shifts right: the cell is less sensitive and needs more hormone for the same response. Part of the reason some drugs lose their effect with heavy use (tolerance).

Up-regulation

Weeks of low hormone make the target cell add receptors. The curve shifts left: the cell is more sensitive.

Dose-response curve

A graph of how strongly a target cell responds at each hormone concentration.

Hormones working together

In your textbook: Silverthorn Chapter 8, the part on hormone interactions.

Dr. Rennie's videos for this part: Synergism and permissiveness.

Synergism

The effect of two or more hormones together is greater than the sum of their separate effects. Example: glucagon, epinephrine and cortisol on blood glucose. Each hormone acts through its own receptor and pathway.

Permissiveness

One hormone cannot exert its full effect unless a second hormone is present, even though the second does little on its own. Example: thyroid hormone and the reproductive hormones in the maturation of the reproductive system.

Functional antagonism

Two hormones have opposite effects through different receptors and pathways. Example: insulin and glucagon on the liver.

Given	Maturation of the reproductive system
Reproductive hormones, no thyroid hormone	Delayed
Thyroid hormone alone	None
Both	Normal

- Thyroid hormone also increases the number of beta receptors in tissues such as the heart, which lets epinephrine have its full effect; too much thyroid hormone can bring a racing heart.
- Synergy among glucagon, epinephrine and cortisol is part of why blood glucose climbs during severe stress or illness.
- Insulin binds a receptor enzyme and tells liver cells to take up glucose and store it as glycogen, so blood glucose falls. Glucagon binds a G protein coupled receptor and tells them to break glycogen down and release glucose, so blood glucose rises.
- Competitive antagonism, two molecules competing for the same receptor, is a different mechanism, used by many drugs.

What tells a gland to release

In your textbook: Silverthorn Chapter 8, the part on control of hormone release.

Dr. Rennie's videos for this part: Endocrine feedback: negative and positive.

OpenStax calls the three triggers humoral, neural and hormonal stimuli. Silverthorn describes the same three as the starting points of endocrine reflex pathways.

Stimulus	What it is	Examples
Humoral	The endocrine cell itself senses a change in the blood	Low blood Ca2+ releases parathyroid hormone (PTH); high blood glucose releases insulin
Neural	A neuron signals the endocrine cell	Sympathetic neurons release acetylcholine onto chromaffin cells of the adrenal medulla, which release epinephrine; the posterior pituitary (next walkthrough)
Hormonal	Another hormone, a trophic hormone, tells the gland to release	TSH from the anterior pituitary releases thyroid hormone (T3 and T4)

- Four small parathyroid glands sit on the back of the thyroid. PTH acts on bone to release Ca2+ into the blood and on the kidneys to keep more Ca2+ out of the urine.
- When the response removes the stimulus, as rising Ca2+ slows PTH release, that is negative feedback.
- The adrenal medulla, the inner part of the adrenal gland, is a modified part of the sympathetic nervous system. Its response comes within seconds.
- Trophic hormone: a hormone that tells another endocrine gland to release its hormone. The next walkthrough follows these chains from the hypothalamus through the pituitary.

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The hypothalamus and pituitary

The written notes for the hypothalamus and pituitary walkthrough. Week 8 competencies 12 to 15.

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Step by step: where the hormones travel

In your textbook: Silverthorn Chapter 8, the parts on the hypothalamus and pituitary gland and on feedback loops.

Dr. Rennie's videos for this part: Thyroid disorders: hyperthyroidism and hypothyroidism.

ADH, from a desert hike to the kidney, step by step

1. You sweat and do not drink, and your plasma osmolarity climbs above the normal of about 290 mOsm/L.
2. Osmoreceptors in the hypothalamus lose a little water, shrink, and fire faster.
3. They excite the ADH neurons, whose cell bodies are in the hypothalamus. ADH was made there and packed into vesicles, which traveled down the axons through the stalk and are stored in the axon terminals in the posterior lobe.
4. Action potentials run down the axons. The terminals release ADH by exocytosis into capillaries in the posterior lobe, and it enters the general circulation.
5. ADH reaches the kidneys and binds V2 receptors on the blood side of the collecting duct cells.
6. Through cAMP, vesicles holding aquaporin-2 water channels move to the urine side of the cell and fuse with the membrane.
7. Water moves by osmosis out of the urine, through the cell, and back into the blood. You make a small amount of dark, concentrated urine.
8. The same rise in osmolarity also makes you thirsty.

Oxytocin in labor, step by step

1. The baby's head presses on the cervix and stretches it.
2. Sensory signals travel to the oxytocin neurons in the hypothalamus.
3. The neurons fire, and oxytocin stored in their terminals in the posterior lobe is released into the blood.
4. Oxytocin makes the smooth muscle of the uterus contract.
5. The contraction pushes the head harder against the cervix, which releases still more oxytocin: positive feedback, until the baby is born.

The cortisol axis, step by step

1. Tier one: neurons in the hypothalamus release corticotropin-releasing hormone (CRH) onto the first capillary bed at the top of the stalk.
2. CRH travels down the portal vessels in the stalk to the second capillary bed in the anterior lobe, still at high concentration.
3. Tier two: CRH makes anterior lobe cells release adrenocorticotropic hormone (ACTH) into the lobe's veins and the general circulation.
4. ACTH reaches the adrenal cortex, the outer layer of the adrenal gland on top of each kidney.
5. Tier three: the adrenal cortex releases cortisol into the general circulation.
6. Long loop: cortisol acts on the hypothalamus and the anterior lobe and lowers CRH and ACTH release.
7. Short loop: ACTH acts on the hypothalamus and lowers CRH release.

The thyroid axis, step by step

1. The hypothalamus releases thyrotropin-releasing hormone (TRH) into the portal vessels.
2. TRH makes the anterior lobe release thyroid-stimulating hormone (TSH) into the general circulation.
3. TSH makes the thyroid gland release thyroid hormone, mostly T4, which tissues convert to the more active T3.
4. Thyroid hormone feeds back on the hypothalamus and the anterior lobe and lowers TRH and TSH release.

Two glands in one

In your textbook: Silverthorn Chapter 8, the part on the hypothalamus and pituitary gland.

Dr. Rennie's videos for this part: The hypothalamus and the pituitary gland, The histology of the pituitary gland.

The hypothalamus is a small region in the floor of the brain that keeps internal conditions steady: water balance, body temperature, hunger and thirst, and the release of many hormones. The pituitary gland hangs from it on a thin stalk and sits in a small cup of bone in the base of the skull.

	Posterior pituitary	Anterior pituitary
Also called	Neurohypophysis	Adenohypophysis
Tissue	Neural tissue: an extension of the hypothalamus	True endocrine tissue: epithelial gland cells
Where it forms before birth	Grows down from the brain	Grows up from the roof of the embryo's mouth
Its hormones	ADH (vasopressin) and oxytocin, made in hypothalamic cell bodies; the lobe stores and releases them	TSH, ACTH, GH, prolactin, FSH and LH, made by its own cells
How the hypothalamus controls it	Directly, through axons that run down the stalk	Through releasing and inhibiting hormones carried in the portal vessels

The posterior pituitary

In your textbook: Silverthorn Chapter 8, the part on the posterior pituitary.

Dr. Rennie's videos for this part: The posterior pituitary and its hormones, The ADH feedback loop in osmoregulation, The second feedback loop for ADH, ADH: what it does and how, ADH: diabetes insipidus and SIADH, Oxytocin in childbirth and bonding.

Neurohormone

A hormone released into the blood by a neuron. ADH and oxytocin are neurohormones.

Antidiuretic hormone (ADH), or vasopressin

A peptide nine amino acids long. Main stimulus: rising plasma osmolarity. Also released when blood pressure or blood volume drops a lot, as after heavy bleeding, sensed by stretch receptors in the heart and blood vessels. Target: the collecting ducts of the kidneys, where it keeps water.

Oxytocin

A peptide nine amino acids long. Stimulus: stretch of the cervix in labor, and suckling at the breast. Targets: the smooth muscle of the uterus, and the cells around the milk glands.

Osmoreceptor

A neuron in the hypothalamus that senses plasma osmolarity. It shrinks a little and fires faster when the plasma gets more concentrated.

Aquaporin-2

The water channel that ADH puts into the urine side of collecting duct cells.

- ADH and oxytocin are made in the cell bodies of two separate groups of hypothalamic neurons, packed into vesicles, carried down the axons through the stalk, and stored in the axon terminals in the posterior lobe. When the neuron fires, they are released by exocytosis into capillaries and enter the general circulation.
- Normal plasma osmolarity is about 290 mOsm/L. Above about 280 mOsm/L, ADH release rises as osmolarity climbs. Rising osmolarity also causes thirst.
- ADH binds the V2 receptor on the blood side of a collecting duct cell. Through cAMP, vesicles carrying aquaporin-2 fuse with the urine side membrane, so water leaves the urine by osmosis and returns to the blood. The blood side of the cell always has water channels.
- High ADH: a small amount of dark, concentrated urine. Low ADH: the channels are pulled back into the cell, and a lot of pale, dilute urine.
- Diabetes insipidus: too little ADH, or kidneys that cannot respond to it. Large amounts of dilute urine and constant thirst.
- Oxytocin in labor contracts the uterus by positive feedback. A synthetic form is given to start or strengthen labor.
- Oxytocin during breastfeeding causes milk ejection (let-down). Making the milk is prolactin's job, from the anterior lobe.
- Both hormones leave the same lobe, but each comes from its own neurons, so the stimulus decides which one is released.

The anterior pituitary

In your textbook: Silverthorn Chapter 8, the parts on the anterior pituitary and the hypothalamic-hypophyseal portal system.

Dr. Rennie's videos for this part: The anterior pituitary and its hormones.

- Neurons in the hypothalamus release releasing hormones and inhibiting hormones onto a first capillary bed at the top of the stalk.
- Portal vessels run down the stalk to a second capillary bed in the anterior lobe. A portal system is any set of vessels that links two capillary beds directly; this one is the hypothalamic-hypophyseal portal system.
- The hypothalamus releases only tiny amounts of these hormones. The portal system delivers them straight to the anterior lobe while their concentration is still high, instead of diluting them in about five liters of blood.

Trophic hormone

A hormone that controls the release of another hormone, usually by another endocrine gland. Most anterior lobe hormones are trophic.

Anterior lobe hormone	Main target	Main control from the hypothalamus
Thyroid-stimulating hormone (TSH)	Thyroid gland	Thyrotropin-releasing hormone (TRH)
Adrenocorticotropic hormone (ACTH)	Adrenal cortex	Corticotropin-releasing hormone (CRH)
Growth hormone (GH)	Liver (which answers by making growth factors) and most other tissues	A releasing hormone, and an inhibiting hormone, somatostatin
Prolactin	Breast: milk production	Held back mainly by dopamine; when dopamine falls, prolactin rises
Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)	Ovaries or testes	Gonadotropin-releasing hormone (covered in the next walkthrough)

All six anterior lobe hormones are peptides or proteins.

Three-tier axes and feedback

In your textbook: Silverthorn Chapter 8, the part on feedback loops and the hypothalamic-pituitary pathways.

Dr. Rennie's videos for this part: ACTH and the stress response, Growth hormone and its feedback, Thyroid anatomy and function, Thyroid hormone synthesis and regulation, T3, T4 and their effects on metabolism.

A three-tier axis is a chain of three hormones: a hypothalamic releasing hormone, an anterior pituitary trophic hormone, and the hormone of a third gland. Only the first one travels in the portal vessels; the other two travel in the general circulation.

Tier	Cortisol axis	Thyroid axis
1. Hypothalamus	CRH	TRH
2. Anterior pituitary	ACTH	TSH
3. Last gland	Adrenal cortex: cortisol	Thyroid gland: T4 and T3
What the last hormone does	Helps the body cope with stress, partly by raising blood glucose	Sets how fast many tissues use energy

Long-loop negative feedback

The last gland's hormone acts on the hypothalamus and the anterior pituitary and lowers their release. It is the main feedback signal in these axes.

Short-loop negative feedback

The pituitary hormone acts on the hypothalamus and lowers releasing hormone release.

- ACTH also keeps the adrenal cortex healthy and full size.
- The thyroid releases mostly T4, and tissues convert it to the more active T3.
- If thyroid hormone rises above normal, TRH and TSH fall. If it falls, TRH and TSH rise.

Primary and secondary disorders

In your textbook: Silverthorn Chapter 8, the part on endocrine pathologies.

Hypersecretion

Too much hormone.

Hyposecretion

Too little hormone.

Primary disorder

The fault is in the last gland of the axis. Through feedback, the trophic hormone moves in the opposite direction to the last gland's hormone.

Secondary disorder

The fault is in the anterior pituitary. The trophic hormone and the last gland's hormone move in the same direction.

Disorder	Cortisol	ACTH	Example
Primary hyposecretion	Low	High	Addison disease (primary adrenal insufficiency): the adrenal cortex is damaged, most often by the immune system; high ACTH can darken the skin
Secondary hyposecretion	Low	Low	Secondary adrenal insufficiency: damage to the ACTH-making cells of the anterior lobe
Primary hypersecretion	High	Low	A tumor of the adrenal cortex that makes cortisol on its own
Secondary hypersecretion	High	High	Cushing disease: a pituitary tumor that makes too much ACTH and ignores feedback

- Cushing syndrome is the picture caused by high cortisol from any cause: high blood glucose, muscle protein breakdown, thin skin, and fat gathered in the face and trunk.
- One lab value, the trophic hormone, separates a primary from a secondary disorder when the last gland's hormone is low in both.
- The same rule works for the thyroid: low thyroid hormone with high TSH is primary hypothyroidism; low thyroid hormone with low TSH is secondary hypothyroidism.

Six months of prednisone, step by step

1. Prednisone acts like cortisol. It is taken in large daily doses for conditions such as severe asthma or rheumatoid arthritis.
2. It feeds back on the hypothalamus and the anterior lobe just as cortisol would, so CRH and ACTH fall very low.
3. Without ACTH, the adrenal cortex makes little cortisol of its own and shrinks over the months.
4. If the drug is stopped suddenly, the axis cannot restart right away, and the shrunken cortex cannot make enough cortisol: adrenal insufficiency, which can be life threatening, especially under stress.
5. So long courses are tapered, lowering the dose slowly while the hypothalamus, pituitary and adrenal cortex recover.

A cut pituitary stalk

- Every anterior lobe hormone except prolactin falls, because the releasing hormones can no longer reach the lobe.
- Prolactin rises, because the dopamine that holds it back is cut off too.
- ADH and oxytocin release from the posterior lobe stops, because the axons that bring them down are cut.

Measuring a hormone

In your textbook: Silverthorn Chapter 8, the parts on endocrine pathologies and on diagnosing endocrine problems.

1. An antibody that grabs only the hormone you want is stuck to the bottom of each well.
2. The sample is added, and the hormone in it is captured.
3. A second antibody carrying an enzyme sticks to the captured hormone.
4. The enzyme turns a colorless chemical colored. The more hormone in the well, the stronger the signal.
5. Wells with known amounts of hormone, the standards, are plotted as signal against amount to make a standard curve.

6. A patient's signal is read across to the curve and down to the amount axis.

- Worked example: a tired, cold patient who is gaining weight has low thyroid hormone. Her TSH reads about 12 mU/L. A typical adult reference range is roughly 0.4 to 4 mU/L, and each lab sets its own, so hers is high: primary hypothyroidism.
- A signal above the highest standard: dilute the sample and run it again, because the curve may not keep going straight past the standards.
- An antibody that also grabs a similar molecule adds that molecule to the signal, so the number is only as good as the antibody.

BIO 005 Human Physiology · Week 8 notes · Dr. Sharilyn Rennie

Reproductive hormones

The written notes for the reproductive hormones walkthrough. Week 8 competencies 16 to 21.

To print: these notes as a two-column PDF. Already printed the Week 8 notes? You do not need to print these again.

How to use these notes. Work through the walkthrough first, writing each prediction on your worksheet before you press Show me. Then read these notes to fill in anything you missed, and use them to review. They are the same facts in a form you can study from, not a copy of the walkthrough.

Step by step: the hormone sequences

In your textbook: Silverthorn Chapter 26, the parts on male reproduction, female reproduction and the menstrual cycle, and pregnancy, labor and lactation.

The male axis, step by step

1. Neurons in the hypothalamus release GnRH in pulses into the portal vessels.
2. GnRH reaches the anterior pituitary, where the gonadotrophs release LH and FSH into the blood.
3. LH reaches the Leydig cells, between the seminiferous tubules, and they make testosterone.
4. Testosterone diffuses into the nearby tubules (a paracrine signal) and also enters the blood.
5. FSH reaches the Sertoli cells inside the tubules. With testosterone, they support spermatogenesis and make androgen-binding protein, which keeps testosterone high around the developing sperm.
6. The Sertoli cells also make inhibin.
7. Testosterone feeds back on the hypothalamus (less GnRH) and the anterior pituitary (less LH). Inhibin feeds back on the anterior pituitary and reduces FSH.

The ovarian cycle, step by step (a 28 day cycle)

1. Day 1 is the first day of menstrual bleeding. FSH rises a little, and a small group of follicles starts to grow.
2. LH drives the theca cells to make androgens. FSH drives the granulosa cells to convert them to estradiol with aromatase.
3. Moderate estradiol, with inhibin, is negative feedback: GnRH, LH and FSH stay low.
4. As FSH falls, one dominant follicle keeps growing and the others break down.
5. The dominant follicle makes more and more estradiol. When estradiol stays high for about two days, its feedback flips to positive.
6. GnRH rises and the pituitary releases the LH surge, with a smaller rise in FSH.
7. The LH surge makes the egg finish its first meiotic division and the follicle rupture. Ovulation happens roughly a day and a half after the surge begins, around day 14.
8. The ruptured follicle becomes the corpus luteum, which makes progesterone, estrogen and inhibin. Feedback is negative again, and LH and FSH stay low.
9. Without a pregnancy, the corpus luteum breaks down after about 12 days. Progesterone and estrogen fall, and the next cycle begins.

The uterine cycle, step by step

1. Menstrual phase: the functional layer is shed with blood, and the basal layer stays.
2. Proliferative phase: estradiol from the growing follicles makes the functional layer thicken, the glands lengthen and the spiral arteries grow up into it. Estrogen also adds progesterone receptors.
3. Secretory phase: after ovulation, progesterone from the corpus luteum makes the glands coil and secrete, and the spiral arteries coil more.
4. When the corpus luteum dies, progesterone falls, the spiral arteries constrict, and the functional layer is starved of oxygen and dies.
5. The arteries relax, blood flows into the dying tissue, and the functional layer is shed. This is day 1 of the next cycle.

Pregnancy, step by step

1. The embryo implants in the lining about a week after ovulation.
2. Its outer cells, the trophoblast, release hCG.
3. hCG binds the LH receptors on the corpus luteum and keeps it alive and making progesterone and estrogen.
4. Progesterone stays high, so the lining is kept and there is no period.
5. Around weeks 7 to 9, the placenta takes over most of the progesterone production, and the corpus luteum is no longer needed.

Labor, step by step (positive feedback)

1. The fetal head presses on the cervix and stretches it.
2. Stretch receptors in the cervix send signals to the hypothalamus.
3. Oxytocin neurons release oxytocin from the posterior pituitary.
4. Oxytocin makes the uterine muscle contract, helped by prostaglandins made in the uterus.
5. Stronger contractions push the head harder into the cervix, so the stretch increases and the loop repeats, stronger each time.
6. Delivery removes the stretch, and the loop stops.

Suckling, step by step

1. The baby suckles, and nerve signals travel from the nipple to the hypothalamus.
2. Prolactin from the anterior pituitary rises (suckling reduces the dopamine that normally holds it down). Prolactin makes the alveolar cells produce milk, mostly for the next feeding.

3. Oxytocin from the posterior pituitary makes the myoepithelial cells around each alveolus contract, and milk is pushed into the ducts: the let-down.

The gonadal axis

In your textbook: Silverthorn Chapter 26, the part on basic patterns of reproduction, on hormonal control.

Dr. Rennie's videos for this part: The gonadotropins: FSH and LH.

Reproduction follows the same hypothalamic-pituitary pattern as the other axes: a releasing hormone from the hypothalamus, a trophic hormone from the anterior pituitary, and a hormone from the target gland that feeds back.

GnRH

Gonadotropin-releasing hormone, from the hypothalamus, carried in the portal vessels to the anterior pituitary.

Gonadotropins

LH (luteinizing hormone) and FSH (follicle-stimulating hormone), released by the gonadotrophs of the anterior pituitary. They act on the gonads, the testes and the ovaries.

Pulsatile GnRH

GnRH is released in bursts. The pituitary needs the pulses: steady GnRH down-regulates its GnRH receptors, and after a brief rise LH and FSH fall. Drugs that act like GnRH, given continuously, are used to shut down the gonads, for example to lower testosterone in some prostate cancers.

Gonad	LH acts on	Which makes	FSH acts on	Which does
Testis	Leydig cells, between the tubules	Testosterone	Sertoli cells, inside the tubules	Supports spermatogenesis; makes androgen-binding protein and inhibin
Ovary	Theca cells, the outer layer of the follicle	Androgens	Granulosa cells, the inner layer	Converts androgens to estradiol (aromatase); makes inhibin

- In both sexes, LH drives the cell that makes the steroid and FSH drives the cell that supports the gamete.
- In both sexes, the FSH-driven cell makes inhibin, which feeds back on FSH alone.

The testis

In your textbook: Silverthorn Chapter 26, the part on male reproduction, on spermatogenesis and its hormonal control.

- Sperm are made in the seminiferous tubules. Spermatogonia sit at the outer wall; spermatocytes and spermatids mature as they move inward; sperm are released into the lumen. This takes a little over two months.
- Sertoli cells run from the wall to the lumen, wrap the germ cells and support them.
- Leydig cells sit outside the tubules, beside blood vessels, and make testosterone.
- Testosterone, a steroid, diffuses into the tubules as a paracrine signal and also enters the blood.
- Androgen-binding protein, made by Sertoli cells under FSH, binds testosterone and keeps it concentrated around the developing sperm.
- Spermatogenesis needs both FSH and testosterone acting on the Sertoli cells.

Blood–testis barrier

Tight junctions between neighboring Sertoli cells, near the outer wall. They separate an outer compartment, open to the blood, holding the spermatogonia, from an inner compartment where later germ cells develop. Developing sperm carry proteins that first appear at puberty, so the barrier keeps immune cells and antibodies from attacking them. It also lets Sertoli cells control the fluid around the sperm.

Hormone	Made by	Feeds back on	Effect
Testosterone	Leydig cells	Hypothalamus and anterior pituitary	Less GnRH and less LH
Inhibin	Sertoli cells	Anterior pituitary	Less FSH

Testosterone from outside

1. High blood testosterone from a shot or anabolic steroids acts as negative feedback.
2. GnRH, LH and FSH fall.
3. Without LH, the Leydig cells stop making testosterone, so the very high level inside the testis falls, even though the blood level is high.
4. Without FSH and local testosterone, Sertoli cells cannot support spermatogenesis. Sperm production falls and the testes can shrink.

The ovarian cycle

In your textbook: Silverthorn Chapter 26, the part on female reproduction and the menstrual cycle, on the ovarian cycle and its hormonal control.

Phase	Days (28 day cycle)	In the ovary	Hormones
Follicular	1 to 14	A group of follicles grows; one becomes dominant	FSH up a little early; estradiol rises; moderate estradiol and inhibin are negative feedback
Ovulation	About 14	The LH surge ruptures the mature follicle and releases the egg	High, sustained estradiol is positive feedback; LH surges, FSH rises a little
Luteal	14 to 28	The corpus luteum forms and, without pregnancy, breaks down after about 12 days into the corpus albicans	Progesterone peaks about a week after ovulation; estradiol has a second, smaller peak; LH and FSH held low

Follicle

An oocyte surrounded by granulosa cells, with a ring of theca cells outside.

Dominant follicle

The one follicle that keeps growing as FSH falls, the one with the most FSH receptors and the best blood supply.

LH surge

A large burst of LH set off when estradiol stays high (for about two days). Positive feedback: high, sustained estradiol makes the hypothalamus and anterior pituitary release more GnRH and LH instead of less. Progesterone, beginning to rise from the follicle, also helps.

Ovulation

Rupture of the follicle and release of the egg, roughly a day and a half (about 32 to 36 hours) after the LH surge begins. The surge also makes the egg finish its first meiotic division. The egg can be fertilized for only about a day.

Corpus luteum

The yellowish gland formed from the ruptured follicle. It makes progesterone, estrogen and inhibin, which hold LH and FSH low so no new follicle matures.

- The surge ends on its own: LH starts turning the follicle cells toward making progesterone, so estradiol falls even before the follicle ruptures, and the pituitary becomes less responsive. After ovulation the follicle is gone.
- The luteal phase is fairly constant at about 14 days because the corpus luteum's lifespan is fixed. The follicular phase is the part that varies. A 32 day cycle ovulates around day 18.
- Counting back 14 days from the next period predicts ovulation better than counting forward from the last one.

Reading the cycle graph

- The LH surge places ovulation: ovulation comes after it.
- The estradiol peak comes about a day before the LH surge and sets it off. It comes from the dominant follicle.
- The progesterone peak, about a week after ovulation, comes from the corpus luteum.
- A progesterone rise in the second half of the cycle proves ovulation happened. In a cycle without ovulation, the LH surge is missing and progesterone never rises.

The uterine cycle

In your textbook: Silverthorn Chapter 26, the part on female reproduction and the menstrual cycle, on the uterine cycle.

Endometrium

The lining of the uterus. The basal layer stays; the functional layer grows each cycle and is shed if no embryo implants. Glands run through it, and spiral arteries supply it.

Myometrium

The smooth muscle of the uterus, under the endometrium.

Uterine phase	Ovarian phase it matches	Driving hormone and source	What the lining does
Menstrual	Start of the follicular phase	Loss of progesterone (corpus luteum dies)	Spiral arteries constrict, the functional layer dies and is shed
Proliferative	Follicular	Estradiol from the growing follicles	Thickens; glands lengthen; spiral arteries grow; progesterone receptors added
Secretory	Luteal	Progesterone (with estrogen) from the corpus luteum	Glands coil and secrete glycogen-rich fluid; arteries coil more; ready for an embryo

- Around ovulation, estrogen makes the cervical mucus thin and watery, which helps sperm through. Progesterone makes it thick and sticky, a plug against bacteria and sperm.
- The lining is lost because progesterone goes away, not because a hormone arrives. If the corpus luteum is rescued and progesterone stays up, the lining stays.

Pregnancy

In your textbook: Silverthorn Chapter 26, the part on pregnancy, on implantation and the hormones of pregnancy.

hCG

Human chorionic gonadotropin, released by the trophoblast (the outer cells of the embryo) after implantation, about a week after ovulation. It is similar in structure to LH and binds the LH receptors already on the corpus luteum, keeping it alive and making progesterone and estrogen. It passes into the urine, where pregnancy tests detect it.

Rescue of the corpus luteum

Because hCG keeps the corpus luteum going, progesterone stays high, the lining is kept and there is no period.

Luteal–placental shift

Around weeks 7 to 9 (sources differ slightly; by the end of the first trimester the placenta is the main source), the placenta takes over most progesterone production. If this failed and the corpus luteum died on schedule, progesterone would fall, the lining would shed, and the pregnancy would be lost.

Hormone	Main source in later pregnancy	Role
hCG	Trophoblast and placenta	Rescues the corpus luteum; peaks in the first trimester, then falls
Progesterone	Placenta	Keeps the uterine muscle quiet and maintains the lining; keeps the breast from making milk
Estrogen	Placenta	Grows the uterus and the breast ducts; late in pregnancy adds oxytocin receptors to the uterine muscle
Relaxin	Corpus luteum and placenta	Thought to soften the cervix and loosen the pelvic ligaments for birth; much of what is known comes from animal studies

Labor and milk

In your textbook: Silverthorn Chapter 26, the part on labor and delivery, and the part on lactation.

- Late in pregnancy, estrogen increases oxytocin receptors on the uterine muscle, and the cervix softens.
- Labor is positive feedback: cervical stretch, then oxytocin, then stronger contractions, then more stretch. Prostaglandins from the uterus add to the contractions.
- A positive feedback loop runs until its stimulus is removed. Delivery removes the stretch and ends the loop.
- Synthetic oxytocin is used to start labor or strengthen contractions.

	Milk production	**Milk ejection (let-down)**
Hormone	Prolactin	Oxytocin
Released from	Anterior pituitary	Posterior pituitary
Acts on	Alveolar cells	Myoepithelial cells around the alveoli
Effect	Make milk, mostly for the next feeding	Contract and push milk into the ducts
Control	Mainly inhibited by dopamine from the hypothalamus; suckling reduces the inhibition	Released by suckling; can become a learned response, such as to a baby's cry; stress can inhibit it

- During pregnancy, estrogen, progesterone and prolactin prepare the breast, but high progesterone keeps it from making milk. After delivery, progesterone falls and milk production begins.
- A mother can make plenty of milk and still not deliver it: production (prolactin) is working, but ejection (oxytocin) is held back, as stress can do.